

5G RAN

Ultra Uplink (High-Frequency TDD) Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue is the first release.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-101240	Ultra Uplink	3.1 Uplink Waveform Adaptation 3.2 Uplink Data Compression 3.3 Full-Power Dual Transmission 3.4 PUSCH Time Domain Enhancement

3 Ultra Uplink

mmWave signal transmission suffers from high path loss and high penetration loss. In addition, mmWave UEs have poor transmission capabilities, and the uplink and downlink coverage is asymmetric, resulting in weak coverage areas. To improve user experience in weak coverage areas, **the Ultra Uplink feature introduces uplink waveform adaptation, uplink data compression, full-power dual transmission, and PUSCH time domain enhancement. These functions improve uplink capabilities, uplink throughput, and uplink user-perceived rate.**

The Ultra Uplink feature is controlled by the **ULTRA_UPLINK_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter. [Table 3-1](#) lists the functions of Ultra Uplink. These functions take effect only after the Ultra Uplink switch is turned on. This feature is supported only in high-frequency TDD.

Table 3-1 Functions of Ultra Uplink

Function Name	Principle Overview	Reference
Uplink waveform adaptation	This function enables a base station to adaptively instruct UEs to switch from the CP-OFDM waveform to the DFT-s-OFDM waveform as the maximum power reduction (MPR) allowed with the DFT-s-OFDM waveform is lower than that allowed with the CP-OFDM waveform. This reduces the amount of power reduction and increases the actual transmit power of UEs.	3.1 Uplink Waveform Adaptation
Uplink data compression	This function uses the 3GPP Deflate compression algorithm to compress duplicate contents in uplink data packets. This reduces resource consumption over the air interface and improves uplink data transmission capability.	3.2 Uplink Data Compression
Full-power dual transmission	This function enables 2T UEs to transmit data at full power when the UEs are working in rank 1 at the cell edge.	3.3 Full-Power Dual Transmission

Function Name	Principle Overview	Reference
PUSCH time domain enhancement	Based on the power delay profile (PDP) of PUSCH DMRS, the base station selects the time-domain filtering coefficient that matches the channel changes based on the time-domain changes of the wanted signal strength. This effectively improves the channel estimation accuracy and uplink coverage.	3.4 PUSCH Time Domain Enhancement

3.1 Uplink Waveform Adaptation

3.1.1 Principles

- The NR system supports Cyclic Prefix-Orthogonal Frequency Division Multiplexing (CP-OFDM) and DFT-spread OFDM (DFT-s-OFDM).
- CP-OFDM allows for the use of non-contiguous frequency-domain resources and flexible resource allocation. It brings a high frequency-diversity gain but produces a high peak to average power ratio (PAPR).
 - DFT-s-OFDM is based on discrete fourier transform (DFT) and requires contiguous frequency-domain resources. It ensures low PAPR (which is similar to that of a single carrier) and high transmit power.
- Uplink waveform adaptation supports adaptation between the CP-OFDM and DFT-s-OFDM waveforms based on the radio environment (such as coverage) of UEs to improve user experience. Uplink waveform adaptation includes RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching.
- RRC-reconfiguration-based dynamic waveform switching: This function is controlled by the **UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter.
 - DCI-based dynamic waveform switching: This function is controlled by the **DYNAMIC_WAVEFORM_SWITCHING_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter.
- When both RRC-reconfiguration-based dynamic waveform switching and DCI-based dynamic waveform switching are enabled, DCI-based dynamic waveform switching preferentially takes effect for UEs capable of dynamic waveform switching, using DCI to indicate the waveform.

 NOTE

In versions earlier than 3GPP R18, the switching between the two waveforms can be implemented only through the *transformPrecoder* IE in the RRCReconfiguration message. For details, see section 6.1.3 "UE procedure for applying transform precoding on PUSCH" in 3GPP TS 38.214 V15.5.0.

According to 3GPP Release 18, the switching between the two waveforms is dynamically indicated by the **Transform precoder indicator** field in the DCI, for details, see section 6.1.3 "UE procedure for applying transform precoding on PUSCH" in 3GPP TS 38.214 V18.1.0 and section 7.3.1 "DCI formats" in 3GPP TS 38.212 V18.1.0.

Before this function takes effect, the network side instructs UEs to always select CP-OFDM.

After this function takes effect, the network side adaptively instructs UEs to select CP-OFDM or DFT-s-OFDM waveform based on the channel quality and the threshold specified by the **NRDUCellPusch.SinrThldforWaveformSel** parameter.

After this function takes effect, if the estimated channel capacity of DFT-s-OFDM is higher than that of CP-OFDM, the uplink SINR is less than the threshold, and single-carrier scheduling is performed on the PUSCH, the network side instructs UEs to select DFT-s-OFDM. In other cases, the network side instructs UEs to select CP-OFDM.

According to section 6.2.2 "UE maximum output power reduction" in 3GPP TS 38.101-2 g50, the maximum power reduction (MPR) of UEs varies with positions of allocated RBs: When the PUSCH waveform remains unchanged, the MPR of a UE to which the inner RB frequency-domain resources are allocated is less than the MPR of a UE to which the outer or edge RB frequency-domain resources are allocated. Therefore, when all the following conditions are met, the spectrum resources at the inner RB position can be preferentially allocated to UEs to obtain the UE power benefits.

- The uplink waveform adaptation function takes effect and the DFT-s-OFDM waveform is used.
- Single-carrier scheduling is performed on the PUSCH, and the carrier bandwidth is 100 MHz, 200 MHz, or 400 MHz.
- The number of UEs to be scheduled in the current TTI is 1.

If the preceding conditions are not met, frequency-domain resources are allocated in the same way as PUSCH time-frequency resource allocation. For details, see *Uplink Scheduling*.

 NOTE

The network side adds a hysteresis range of ± 2 dB to the threshold to prevent ping-pong waveform switching.

The frequency-domain positions of inner RBs, outer RBs, and edge RBs are related to the configured carrier bandwidth, number of RBs scheduled, and UE power class. For details, see sections 6.2.2 "UE maximum output power reduction" and 6.2A.2 "UE maximum output power reduction for CA" in 3GPP TS 38.101-2 g50.

3.1.2 Network Analysis

3.1.2.1 Benefits

This function increases the UE transmit power and can increase the average uplink UE throughput when the DFT-s-OFDM waveform is selected and inner RB frequency-domain resources are scheduled.

This function does not produce any gains in single-UE multi-stream transmission scenarios, where DFT-s-OFDM is not applicable and the UE still works with CP-OFDM.

DCI-based dynamic waveform switching uses DCI to indicate waveform switching, reducing RRC signaling overhead and mitigating the risk of service drops introduced by RRCReconfiguration.

NOTE

The 3GPP-defined MPR is the upper limit of the maximum power reduction value for UEs. The MPRs of UEs provided by different vendors on the live network meet the protocol requirements as long as their values are less than or equal to this upper limit. As such, the transmit power of different UEs on the live network may differ, so do the benefits delivered after this function takes effect.

For high-frequency cells, it is recommended that the switches for uplink waveform adaptation be configured consistently across all carriers within the same sector. Otherwise, uplink waveform adaptation may not take effect, resulting in no or reduced benefits.

3.1.2.2 Impacts

The impacts between this function and other functions are described in [3.1.3.2.3 Function Impacts](#).

CP-OFDM can use non-contiguous frequency-domain resources while DFT-s-OFDM can use only contiguous frequency-domain resources. If UEs select DFT-s-OFDM rather than CP-OFDM in non-contiguous frequency-domain scenarios, the scheduling bandwidth is smaller, and the single-UE throughput may not increase or may decrease. The specific negative results depend on the changes in (1) scheduled bandwidth and power spectral density and (2) increase in UE transmit power after the waveform mode is switched from CP-OFDM to DFT-s-OFDM.

After uplink waveform adaptation takes effect, the maximum transmit power of a UE changes abruptly when the UE switches between CP-OFDM and DFT-s-OFDM. Specifically, the UE throughput increases significantly when the UE switches from CP-OFDM to DFT-s-OFDM, and conversely, decreases significantly when the UE switches from DFT-s-OFDM to CP-OFDM.

After the uplink waveform adaptation function is enabled, compatibility issues may occur on UEs not supporting the DFT-s-OFDM waveform, causing service drops.

3.1.3 Requirements

3.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

3.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_ULINK_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	Uplink waveform adaptation can take effect only if Ultra Uplink is enabled.

3.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Uplink DMRS Type	NRDUCellPusch.UlDMRSType	<i>Uplink Scheduling</i>	When NRDUCellPusch.UlDMRSType is set to TYPE2 in a cell, the DYNAMIC_WAVEFORM_SWITCHING_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter must be deselected.

3.1.3.2.3 Function Impacts

None

3.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

3.1.3.4 Cells

DCI-based dynamic waveform switching can be enabled only when the **NrDuCell.HighSpeedFlag** parameter is not set to **HIGH_SPEED** and the **NrDuCell.NrDuCellNetworkingMode** parameter is set to **NORMAL_CELL**.

3.1.3.5 Others

DCI-based dynamic waveform switching requires that UEs report their dynamic waveform switching capabilities through the **dynamicWaveformSwitch-r18** and **dynamicWaveformSwitchIntraCA-r18** fields.

3.1.4 Operation and Maintenance

3.1.4.1 Data Configuration

3.1.4.1.1 Data Preparation

[Table 3-2](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 3-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	ULTRA_UPLINK_SW	Select this option if the Ultra Uplink feature is required.
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_WAVEFORM_ADAPT_SW	It is recommended that this option be selected in scenarios with limited coverage and be deselected in other scenarios.
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	DYNAMIC_WAVEFORM_SWITCHING_SW	It is recommended that this option be selected in scenarios with limited uplink coverage and be deselected in other scenarios.
SINR Threshold for Waveform Selection	NRDUCellPuschSinrThldforWaveformSel	None	Use the default value.

3.1.4.1.2 Using MML Commands

Before using MML commands, refer to [3.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Ultra Uplink
//Enabling Ultra Uplink
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UlCoverageAlgoSwitch=ULTRA_UPLINK_SW-1;
- Uplink waveform adaptation
//Enabling RRC-reconfiguration-based dynamic waveform switching
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL_WAVEFORM_ADAPT_SW-1;
//Enabling DCI-based dynamic waveform switching
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,

```
UICoverageAlgoSwitch=DYNAMIC_WAVEFORM_SWITCHING_SW-1;  
//Setting the SINR threshold for waveform selection  
MOD NRDUCELLPUSCH: NrDuCellId=0, SinrThldforWaveformSel=32;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Uplink waveform adaptation
//Disabling RRC-reconfiguration-based dynamic waveform switching
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
AdaptiveEdgeExpEnhSwitch=UL_WAVEFORM_ADAPT_SW-0;
//Disabling DCI-based dynamic waveform switching
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UICoverageAlgoSwitch=DYNAMIC_WAVEFORM_SWITCHING_SW-0;
- Ultra Uplink
//Disabling Ultra Uplink
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=ULTRA_UPLINK_SW-0;

3.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.1.4.2 Activation Verification

Check the values of **N.ChMeas.RB.PUSCH.64QAMTP.RANK1.MCS.0** to **N.ChMeas.RB.PUSCH.64QAMTP.RANK1.MCS.27** (number of allocated RBs with rank 1 and an MCS index selected for scheduling) in the function subset ChMeas.PUSCH.64QAMTP.MCS.DuCell. If the values are all zeros before this function is enabled but some turn into non-zero ones after this function is enabled, this function has taken effect.

3.1.4.3 Network Monitoring

After this function is enabled, monitor its running status by checking the increases in **User Uplink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**.

3.2 Uplink Data Compression

3.2.1 Principles

Principles of Uplink Data Compression

Uplink data compression (also referred to as UDC) uses the 3GPP Deflate compression algorithm to compress uplink data packets. This reduces resource consumption over the air interface and improves uplink data transmission

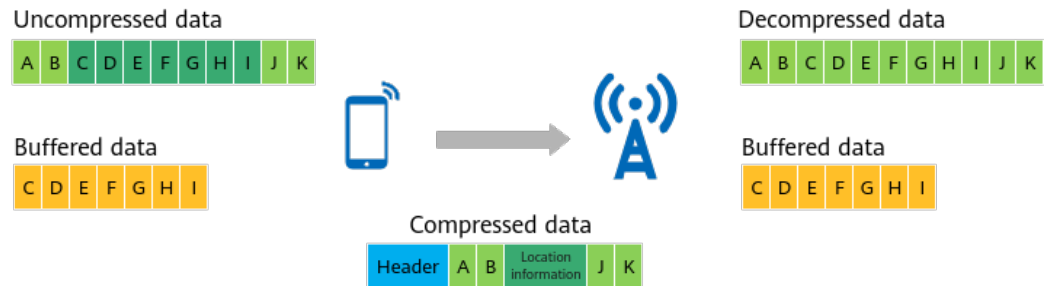
capability. This function applies to the PDCP layers of the UE and gNodeB. After this function is enabled, the UE compresses uplink data packets that are not compressed or encrypted and then the gNodeB decompresses the compressed packets to reduce the amount of data transmitted over the air interface. This function is illustrated in [Figure 3-1](#) and its principles are described as follows:

1. The UE and the gNodeB each maintain a buffer with the same size and the same data.
2. Compression on the UE side
 - If the to-be-transmitted data includes content that is the same as the data in the buffer, the content is not transmitted; instead, the location information of the content in the buffer is transmitted.
 - If the to-be-transmitted data has content different from the data in the buffer, the content is directly transmitted.

A UDC packet header is used to indicate whether data is compressed. The final compressed data consists of the UDC packet header and the content that is different from the buffered data.

3. Decompression on the gNodeB side
After receiving the data, the gNodeB determines whether the data has been compressed based on the packet header. If the data has been compressed, the gNodeB reads the duplicate content from the buffer based on the location information and reassembles the duplicate content with the non-duplicate content to restore the original data.

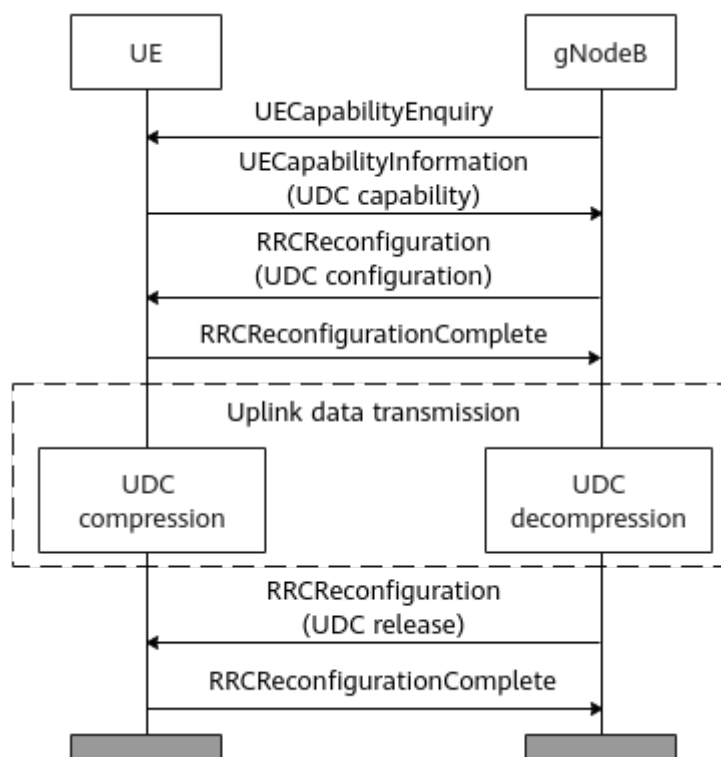
Figure 3-1 Principles of uplink data compression



Uplink Data Compression Procedure

[Figure 3-2](#) shows the uplink data compression procedure.

Figure 3-2 Uplink data compression procedure



1. The UE reports its UDC capability through the **udc-r17** field.
2. The UE and the gNodeB perform the following operations after completing the setup of an RRC connection:
 - a. The gNodeB sends an **RRCReconfiguration** message to the UE, instructing the UE to enable uplink data compression.
 - b. The UE responds to the gNodeB with an **RRCReconfigurationComplete** message.
3. The UE compresses uplink data packets, and the gNodeB decompresses the compressed packets.
4. The gNodeB disables uplink data compression and then performs the following operations:
 - a. The gNodeB sends an **RRCReconfiguration** message to the UE, instructing the UE to disable uplink data compression.
 - b. The UE responds to the gNodeB with an **RRCReconfigurationComplete** message.

Parameters Related to Uplink Data Compression

Uplink data compression can be configured at the cell level or QCI level.

- Cell-level parameter configuration
 - **UDC_SW** option of the **NRDUCellChnCovAlgo.UICoverageAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for all uplink data packets used for data transmission at the NAS layer in a cell.

- **UPLINK_SIG_COMPRESS_SW** option of the **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** parameter: If this option is selected, uplink data compression can take effect for all uplink data packets used for signaling transmission at the NAS layer in a cell.
- **NRCeIIServExp.UdcBufferSize**: This parameter specifies the buffer size on the gNodeB side. The actual buffer size is equal to the value of this parameter or the buffer size reported by the UE, whichever is smaller. A smaller buffer size results in a lower compression efficiency but a shorter decompression delay. A larger buffer size produces the opposite effects.
- QCI-level parameter configuration
QCI_UDC_SW option of the **gNBQciBearer.PdcpAlgoSwitch** parameter: This option specifies whether to enable QCI-specific uplink data compression. If this option is selected, uplink data compression can take effect for bearers with the specified QCI.
 - For such bearers used for data transmission at the NAS layer, if the **UDC_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.
 - For such bearers used for signaling transmission at the NAS layer, if the **UPLINK_SIG_COMPRESS_SW** option of the cell-level parameter **NRDUCellChnCovAlgo.UlCoverageAlgoSwitch** is selected, this cell-level parameter setting takes effect; otherwise, the QCI-level parameter setting takes effect.

A maximum of two uplink DRBs can be compressed for a single UE. This function is supported only in SA networking.

3.2.2 Network Analysis

3.2.2.1 Benefits

This function improves the compression efficiency of uplink data packets, decreases the amount of data transmitted over the air interface, and reduces the transmission delay for CEUs. The higher the compression efficiency of uplink data packets, the greater the decrease in the amount of data transmitted over the air interface.

NOTE

Compression efficiency = $(1 - \frac{\text{N.PDCP.Vol.Ul.UDC.BefDecomp}}{\text{N.PDCP.Vol.Ul.UDC.AftDecomp}}) \times 100\%$

3.2.2.2 Impacts

The impacts between this function and other functions are described in [3.2.3.2.3 Function Impacts](#).

After this function takes effect:

- Compressed data packets carry compression indication information, causing extra overhead. Therefore, the size of data packets after compression may be

greater than that before compression and there may be negative impacts if the repetition rate of uplink data is not high, for example, if uplink data packets have been compressed or encrypted.

- The average gNodeB board user-plane CPU usage (indicated by **VS.NRBoard.UPlane.CPULoad.Avg**) increases, the uplink PRB usage (indicated by **Uplink Resource Block Utilizing Rate (DU)**) may decrease, and the uplink MCS index fluctuates.

3.2.3 Requirements

3.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

3.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_UPLINK_SW option of the NRDUCellConfigCovAlgo.UlCoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	Uplink data compression can take effect only if Ultra Uplink is enabled.

3.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	QoS Class Identifier	gNBQciBearer.Qci	<i>QoS Management</i>	When the gNBQciBearer.Qci parameter is set to 1 , 65 , or 66 , the QCI_UDC_SW option of the gNBQciBearer.PdcpAlgorithm parameter cannot be selected.

3.2.3.2.3 Function Impacts

None

3.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

3.2.3.4 Networking

This function is supported only in SA networking.

3.2.3.5 Others

This function requires that UEs report their UDC capabilities through the **udc-r17** field.

3.2.4 Operation and Maintenance

3.2.4.1 Data Configuration

3.2.4.1.1 Data Preparation

Table 3-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 3-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	ULTRA_UPLINK_SW	Select this option if the Ultra Uplink feature is required.
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	UDC_SW	Select this option.
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	UPLINK_SIG_COMPRESS_SW	Select this option.
UDC Buffer Size	NRCellServExp.UdcBufferSize	None	Use the recommended value.
PDCP Algorithm Switch	gNBQciBearer.PdcpAlgoSwitch	QCI_UDC_SW	If uplink data compression is required only for uplink data packets carried on bearers with the specified QCI, deselect the UDC_SW or UPLINK_SIG_COMPRESS_SW option of the NRDUCellChnCovAlgo.UlCoverageAlgoSwitch parameter and select the QCI_UDC_SW option for the corresponding QCI.

3.2.4.1.2 Using MML Commands

Before using MML commands, refer to [3.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- Ultra Uplink
//Enabling Ultra Uplink
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=ULTRA_UPLINK_SW-1;
- Cell-level uplink data compression
//Selecting the UDC_SW option if uplink data compression is required for all uplink data packets used for data transmission at the NAS layer in a cell
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UDC_SW-1;
//Selecting the UPLINK_SIG_COMPRESS_SW option if uplink data compression is required for all uplink data packets used for data transmission at the NAS layer in a cell
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UPLINK_SIG_COMPRESS_SW-1;
- QCI-specific uplink data compression
//Deselecting the UDC_SW and UPLINK_SIG_COMPRESS_SW options and selecting the QCI_UDC_SW option if uplink data compression is required only for uplink data packets carried on bearers with the specified QCI (for example, QCI 8)
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UDC_SW-0;
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0,
UICoverageAlgoSwitch=UPLINK_SIG_COMPRESS_SW-0;
MOD GNBQCIBEARER: Qci=8, PdcAlgoSwitch= QCI_UDC_SW-1;
- Other parameters
//Configuring the buffer size for uplink data compression
MOD NRCELLSERVEXP: NrCellId=0, UdcBufferSize=8KBYTE;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Cell-level uplink data compression
//Deselecting the UDC_SW option
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UDC_SW-0;
//Deselecting the UPLINK_SIG_COMPRESS_SW option
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=UPLINK_SIG_COMPRESS_SW-0;
- QCI-specific uplink data compression
//Deselecting the QCI_UDC_SW option for the corresponding QCI
MOD GNBQCIBEARER: Qci=8, PdcAlgoSwitch= QCI_UDC_SW-0;
- Ultra Uplink
//Disabling Ultra Uplink
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UICoverageAlgoSwitch=ULTRA_UPLINK_SW-0;

3.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.2.4.2 Activation Verification

This function has taken effect if the compression efficiency of uplink data packets increases after this function is enabled.

Compression efficiency = $(1 - \text{N.PDCP.Vol.Ul.UDC.BefDecomp} / \text{N.PDCP.Vol.Ul.UDC.AftDecomp}) \times 100\%$

3.2.4.3 Network Monitoring

After this function is enabled, you can monitor the running status of this function by observing the increase in compression efficiency.

3.3 Full-Power Dual Transmission

3.3.1 Principles

Full-power dual transmission is introduced to improve the performance of CEUs. **This function allows UEs to transmit at full power** and achieve better performance **when the UEs work in rank 1 and have two or more transmit antennas and SRS ports**. This function is controlled by the **UL_RANK1_FULL_POWER_SW** option of the **NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch** parameter. After full-power dual transmission is enabled, it works as follows:

1. The base station checks the *ul-FullPwrMode-r16*, *ul-FullPwrMode1-r16*, and *ul-FullPwrMode2-TPMIGroup-r16* fields in the *FeatureSetUplink* IE to determine whether a UE supports full-power dual transmission. For details, see [3.3.3.4 Others](#).
2. If the UE supports this function, the base station sets the *ul-FullPowerTransmission-r16* IE in *PUSCH-Config*.
3. After the UE completes configuration, full-power dual transmission takes effect.

NOTE

For details about the *ul-FullPwrMode-r16*, *ul-FullPwrMode1-r16*, and *ul-FullPwrMode2-TPMIGroup-r16* fields, see section 4.2.7.7 "FeatureSetUplink parameters" in 3GPP TS 38.306-g50.

3.3.2 Network Analysis

3.3.2.1 Benefits

This function increases the uplink transmit power and uplink user-perceived rate at the coverage edge if UEs support full-power transmission in the uplink and work in rank 1.

NOTE

To ensure safety, the transmit power of a UE must be lower than a specified SAR when the UE is close to the human body. Therefore, to achieve the function gain, it is required that the transmit power of the UE not be limited due to the specified SAR.

3.3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.3.2.3 Function Impacts](#).

After this function takes effect, the transmit power of CEUs increases, slightly increasing interference.

3.3.3 Requirements

3.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

3.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_UPLINK_SW option of the NRDUCellC hnCovAlgo. UICoverage AlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	Full-power dual transmission can take effect only if Ultra Uplink is enabled.

3.3.3.2.2 Mutually Exclusive Functions

None

3.3.3.2.3 Function Impacts

None

3.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

3.3.3.4 Others

This function requires that UEs support one of the following transmission modes: FullPower, FullPowerMode2, and FullPowerMode1. Whether a UE supports the three transmission modes is determined by the following capability fields in the *FeatureSetUplink* IE: ul-FullPwrMode-r16, ul-FullPwrMode2-TPMIGroup-r16, and ul-FullPwrMode1-r16. If the UE reports two or more capability fields at the same time, the base station instructs the UE to use the transmission modes in the following priorities: FullPower > FullPowerMode2 > FullPowerMode1.

3.3.4 Operation and Maintenance

3.3.4.1 Data Configuration

3.3.4.1.1 Data Preparation

[Table 3-4](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 3-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	ULTRA_UPLINK_SW	Select this option if the Ultra Uplink feature is required.
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	UL_RANK1_FULL_POWER_SW	- It is recommended that this option be selected if coverage is limited. - It is recommended that this option be deselected in other scenarios.

3.3.4.1.2 Using MML Commands

Before using MML commands, refer to [3.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Ultra Uplink
//Enabling Ultra Uplink
MOD NRDUCCELLCHNCOVALGO: NrDuCellId=0, UlCoverageAlgoSwitch=ULTRA_UPLINK_SW-1;
- Full-power dual transmission
//Enabling full-power dual transmission
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL_RANK1_FULL_POWER_SW-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Full-power dual transmission
//Disabling full-power dual transmission
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=UL_RANK1_FULL_POWER_SW-0;
- Ultra Uplink
//Disabling Ultra Uplink
MOD NRDUCCELLCHNCOVALGO: NrDuCellId=0, UlCoverageAlgoSwitch=ULTRA_UPLINK_SW-0;

3.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.3.4.2 Activation Verification

If the uplink DMRS RSRP increases after this function is enabled, this function has taken effect.

3.3.4.3 Network Monitoring

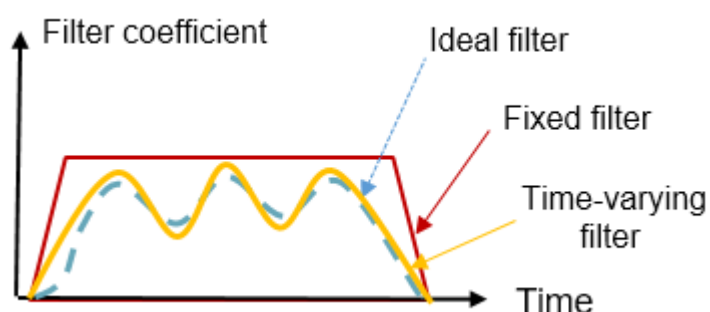
After this function is enabled, monitor its running status by checking the increases in **User Uplink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**.

3.4 PUSCH Time Domain Enhancement

3.4.1 Principles

The complexity of the radio propagation environment means that signals arrive at the receiver after a series of time-variant fading and delay. A simple fixed-coefficient channel filter design cannot track channel changes in a time-varying manner, impairing performance as a result. As such, the PUSCH time domain enhancement function is introduced. Based on the power delay profile (PDP) of PUSCH DMRS, the base station selects the time-domain filtering coefficient that matches the channel changes based on the time-domain changes of the wanted signal strength, as shown in [Figure 3-3](#). This effectively improves the channel estimation accuracy and uplink coverage. This function is controlled by the **PUSCH_TIME_DOMAIN_ENH_SW** option of the **NRDUCellAlgoSwitch**. *FullChannelCovEnhSwitch* parameter.

Figure 3-3 Principles of PUSCH time domain enhancement



3.4.2 Network Analysis

3.4.2.1 Benefits

PUSCH time domain enhancement increases the **User Uplink Average Throughput (DU)** when QPSK is used for uplink scheduling.

3.4.2.2 Impacts

The impacts between this function and other functions are described in [3.4.3.2.3 Function Impacts](#).

This function has no impact on the network.

3.4.3 Requirements

3.4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-101240	Ultra Uplink	NR0S00ULUL00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

3.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Ultra Uplink	ULTRA_ULINK_SW option of the NRDUCellConfigCovAlgo.UplinkCoverageAlgoSwitch parameter	<i>Ultra Uplink (High-Frequency TDD)</i>	PUSCH time domain enhancement can be enabled only if the Ultra Uplink switch is turned on.

3.4.3.2.2 Mutually Exclusive Functions

None

3.4.3.2.3 Function Impacts

None

3.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

5900 series base stations (macro base stations)

Boards

All NR-capable main control boards and NR TDD mmWave baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

3.4.3.4 Cells

PUSCH time domain enhancement has the following requirements:

- The **NRDUCellTrp.TxRxMode** parameter must be set to 4T4R or 8T8R for NR cells.
- The **NRDUCell.SlotAssignment** parameter must be set to **4_1_DDDSU** for NR cells.

3.4.4 Operation and Maintenance

3.4.4.1 Data Configuration

3.4.4.1.1 Data Preparation

[Table 3-5](#) describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 3-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UL Coverage Algorithm Switch	NRDUCellChnCovAlgo.UlCoverageAlgoSwitch	ULTRA_UPLINK_SW	Select this option if the Ultra Uplink feature is required.
Full Channel Coverage Enhancement Switch	NRDUCellAlgoSwitch.FullChannelCovEnhSwitch	PUSCH_TIME_DOMAIN_ENH_SW	Select this option for cells in which uplink coverage needs to be improved.

3.4.4.1.2 Using MML Commands

Before using MML commands, refer to [3.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Ultra Uplink
//Enabling Ultra Uplink
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UlCoverageAlgoSwitch=ULTRA_UPLINK_SW-1;
- PUSCH time domain enhancement
//Enabling PUSCH time domain enhancement
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FullChannelCovEnhSwitch=PUSCH_TIME_DOMAIN_ENH_SW-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- PUSCH time domain enhancement
//Disabling PUSCH time domain enhancement
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, FullChannelCovEnhSwitch=PUSCH_TIME_DOMAIN_ENH_SW-0;
- Ultra Uplink
//(Optional) Disabling Ultra Uplink
MOD NRDUCELLCHNCOVALGO: NrDuCellId=0, UlCoverageAlgoSwitch=ULTRA_UPLINK_SW-0;

3.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.4.2 Activation Verification

You can check whether this function has taken effect by observing whether the **User Uplink Average Throughput (DU)** increases.

3.4.4.3 Network Monitoring

After this function is enabled, monitor its running status by checking the increases in **User Uplink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**.

4 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.212: "NR; Multiplexing and channel coding"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- *Uplink Scheduling*
- *Turbo Uplink (Low-Frequency TDD)*
- *QoS Management*
- *License Control Item Lists*